Impact Of Climate Changes On Overhead Transmission Lines

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***Abstract*—The consequences of climate change are felt everywhere, especially in power systems and overhead lines as one of its most exposed elements. It is expected that impact of climate changes will be even more expressed in the coming decades, with an increasing possibility of occurrence of extreme conditions in the form of storms, winds, snow, ice, high temperatures, fires, etc. Therefore, it is essential to ensure the safe and stable operation of power networks. In this paper, an experience with two strong storms which appeared within only two days in July, 2023 affecting larger part of Croatian transmission network is presented, in order to discuss whether it is necessary to change valid requirements at design phase of new overhead lines. Also, it shows a huge significance of neighbouring transmission system collaboration, especially in terms of modular emergency restoration tower exchange in cases of unpredictable weather phenomena.**

Keywords—climate changes, storm, wind, ice, transmission network, OHL, design, modular towers, TSO collaboration

I. INTRODUCTION

Over the past decade, climate change has manifested in various forms, significantly affecting power system infrastructure. The escalating frequency and intensity of extreme weather events, such as hurricanes, wildfires, floods, and heatwaves, have posed unprecedented challenges to the resilience and reliability of power grids. Rising global temperatures have led to increased energy demand for cooling, placing additional stress on power generation and distribution systems. Moreover, changing precipitation patterns affect hydropower generation, a key component of many energy portfolios.

These climate-induced disruptions have resulted in widespread power outages, economic losses, and compromised energy security. Infrastructure damage, prolonged downtime, and repair costs have escalated, prompting a re-evaluation of traditional approaches to power system design and maintenance. In response, there is a growing emphasis on implementing climate-resilient technologies, enhancing grid flexibility, and diversifying energy sources to mitigate the impact of climate change on power systems. Policymakers, utilities, and industry stakeholders are now recognizing the urgency of adopting sustainable practices and investing in innovative solutions to build a more robust and adaptable power infrastructure for the future.

Croatia is Mediterranean country, located in Southeast Europe. Due to its geographical position, Croatian power system comprises plants and facilities for electricity production, transmission and distribution located in different climatic zones and geographical areas. The transmission network is a part of the Croatian power system and comprises transformer substations, switchyards, overhead lines (OHLs) and underground cables. Croatian Transmission System Operator (further in text HOPS) owns 7778 kilometres of overhead lines 400 kV, 220 kV and 110 kV voltage levels and 187 substations.

Scientific research, showed that climate change has a far greater impact in the Mediterranean area than in other parts of the world, and as such Croatia is subjected to the most extreme climate changes, and like in many other regions, are multifaceted and can affect various aspects of the environment, economy, and society. Some notable impacts of climate change in Croatia also are visible in energy sector. The frequency and intensity of extreme weather events, such as heatwaves, storms, and wildfires, have increased. These events can have direct and indirect impacts. Currently, there are few papers and studies on the impact of climate changes on the power systems in the scientific and technical literature [1-5].

The first part of the paper present HOPS experiences with two strong storms, that occurred in July 2023 (on 19th and 21st) on the territory of eastern Croatia and caused unprecedented damage on the transmission network of HOPS, whereby in total 51 transmission line towers were damaged or completely demolished (including six cross-border OHLs which enable interconnection to Hungary, Serbia and Bosnia Herzegovina).

For the emergency repair of the transmission network and to ensure reliable electricity supply, a modular emergency restoration towers were installed, with coordination and help provided from neighbouring TSOs, which borrowed their modular emergency towers.

It is important to mention that over the past few years HOPS experienced several damages on its network, as a direct result of climate factors, including earthquakes, floods, fires, devastating storms, lightning strikes etc.

Similar events with significant impact on HOPS network were also recorded earlier, for example in February 2014, when ice caused significant damage to the network, in August 2018 when strong wind damage severely towers and completely demolished one tower on 110 kV OHL [6], or in August 2021 (wind) when three towers of internal 400 kV OHL in row were completely demolished [7]. The network is also often affected with salt and fires in coastal areas, or wind, ice and floods in continental areas. Also, as a result of the devastating earthquakes in 2020, significant damage occurred in 7 substations in the wider Zagreb and Sisak area.

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Second part of the paper gives an overview of existing national norms and standards related to the design and engineering of OHLs (in this case, parameters related to wind), and highlight some recommendations in order to consider whether changing of those requirements could prevent future damages and OHL outages do unpredictable weather phenomena.

II. Impact of Climate Changes on OHLs in Croatian Transmission Network

Croatian transmission network consists of four different areas, based on four main and biggest cities - transmissions area Zagreb (TA ZG), Split (TA ST), Rijeka (TA RI) and Osijek (TA OS). Each of them is located in different geographical zone with different climatic conditions, and possible impact on power system equipment.

Continental area (parts of TAs ZG and OS), especially in winter, can experience heavy snowfall and ice. Accumulation of snow and ice on overhead lines and towers or substation equipment may lead to increased weight, causing sagging, and potential damage. Ice accretion can also increase the risk of conductor breakage. Also, strong winter winds can exacerbate the impact of snow and ice by causing additional stress on power lines.

Area of Adriatic Coast (parts of TAs ST and RI) is exposed to strong winds and salt-laden air. Salt corrosion is a significant concern for power system infrastructure, as it can accelerate the deterioration of materials, leading to equipment failure and increased maintenance needs. Also, storm surges associated with coastal storms can lead to flooding, impacting substation equipment and potentially causing electrical failures. In summer, coastal regions may also face the risk of wildfires.

Higher elevations (parts of TAs ST and RI), in mountainous regions receive heavy snowfall during winter. Snow accumulation can affect power lines and transformers, and ice formation on elevated structures poses a risk of conductor breakage and equipment damage.

And, at the end, low-lying areas and river valleys may be susceptible to flooding during periods of heavy rainfall, which can damage substation equipment, transformers, and other infrastructure, leading to extended outages.

Understanding these regional climate-related challenges is crucial for power system operators and planners. Mitigation strategies include designing infrastructure to withstand specific environmental conditions, implementing regular maintenance and inspection programs, and employing materials that are resistant to corrosion and environmental stress. Additionally, advanced forecasting and monitoring systems can help anticipate extreme weather events and facilitate timely response and preventive measures to enhance the resilience of the power grid.

On Wednesday, July 19th, 2023, north-western and eastern Croatia was hit by a strong storm accompanied by hurricane-force winds and thunder, which caused a serious breakdown in the power system [8]. The disaster had a significant impact on the power system, causing damage to transmission lines, collapse of poles and consequent interruptions in supply, requiring immediate interventions and infrastructure restoration to normalize the power system in the affected areas of TAs ZG and OS. In addition, on July 21st, 2023, another storm further worsened the situation, contributing to new transmission line outages in the area of the transmission network of Zagreb (and city Kutina).

The storm that engulfed the plain area was extremely strong, and resulted in its transformation into a supercell. This storm caused serious material damage, including hail with a diameter of about 10 centimetres, extremely strong wind shock fronts and numerous damaged buildings in the area of Zagreb and Slavonia. The highest recorded wind gust reached an exceptional 52.3 m/s (188 km/h) in city Kutina, which significantly exceeded the designed endurance of the transmission line.

During the afternoon, a convective storm formed west of Slovenia and intensified around the border area of Italy, Slovenia and Austria. The storm moved towards the affected areas from the west (Slovenia) towards Zagreb and further east. Radar images (source: State Hydrometeorological Institute - DHMZ) show the direction of the storm from 15:30 (immediately before the first driving event) until 18:30, in steps of 30 minutes (Figure 1).

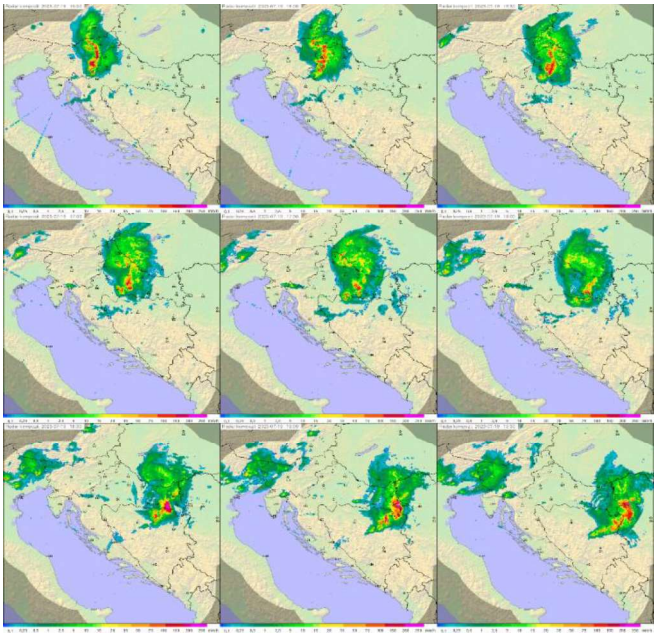


Fig. 1 - The storm over Croatia from 15:30-18:30 h on radar

On its way to the east, the storm came over a plain area with ground temperatures of 28-31 °C, and dew points around 20 °C, which enabled further strengthening in a CAPE (Convective Available Potential Energy) environment of over 1500 J/kg, in places even more than 2500 J/kg. During its rapid movement to the east, the storm system took on the characteristics of a supercell, and produced localized hail with a diameter of about 10 centimetres.

In addition to the hail, the storm was characterized by an extremely strong wind shock front (called downburst), which, according to measurements, exceeded 20-30 meters per second at the largest number of affected measurement locations. The highest wind gust was recorded at 17:04, in the area of city Kutina and was 52.3 m/s (188 km/h), as it is shown on Figure 2, what is a significantly stronger gust than the OHLs were designed for (described in next chapter).

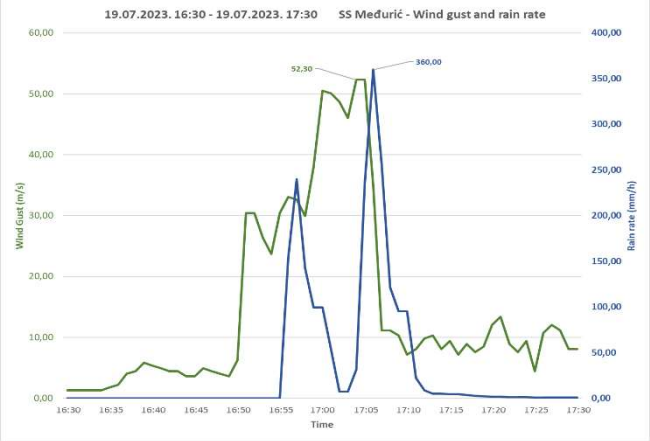


Fig. 2 - Wind gusts during 19.07.2023. at the SS Međurić equipped with automatic measurement system

In parallel with ongoing storm, numerous lightning strikes occurred on mentioned areas of TAs ZG and OS, causing direct outages of nine overhead lines (Figure 3). HOPS uses system for locating atmospheric discharges (SLAP), that provides data on lightning strikes that occurred over the observed area. The SLAP system is connected to the SCADA system and enables the correlation of lightning strikes with the indication of events in the network.

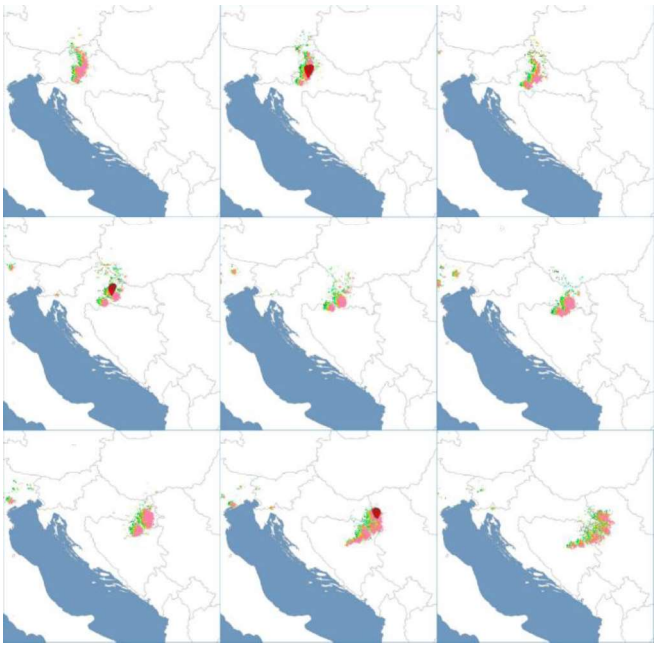


Fig. 3 – Lightning strikes recorded during storm with lightning location system SLAP

An average wind speed during storm on July 19th, 2023, was 88 km/h and the storm encompassed 350 km wide area, whereby system SLAP recorded 363.872 lightning strikes (from 14:00 until 20:00), what equals 1010 strikes / min, with 9 OHL outages.

Similarly, an average wind speed during storm on July 21st, 2023, was 94 km/h and the storm encompassed 470 km wide area, whereby system SLAP recorded 718.184 lightning strikes (from 14:00 until 20:00), what equals 1994 strikes / min, with 4 OHL outages.

As a consequence, due to the effects of the storm, outage occurred on 14 overhead lines in the TA ZG and 8 in the TA OS with significant damage of towers and other equipment.

In the TA ZG, a total of 20 towers were damaged, 13 of which were broken, that is, the damage to the structure is such that it is not possible to correct the damage only with locksmith interventions. Looking at the km of the route, a little more than 5 km of the transmission line route was damaged in total. For the needs of temporary transmission line operation, 10 modular emergency towers were used.

In the TA OS, a total of 31 towers were damaged. Of these, 27 were completely demolished and 4 towers were damaged. Looking at the kilometres of the route, a total of about 12 km of the transmission line route was damaged. For the needs of the temporary operation of the transmission line, 12 modular emergency towers were installed. On the 110 kV voltage level transmission lines, modular towers were installed by HOPS workers, and on the interconnection (HR-RS) OHL 400 kV Ernestinovo - Sremska Mitrovica, two emergency towers were installed by the workers of an external contractor. Overview of consequences are shown in Table 1 [8].

TABLE I. OVERVIEW OF THE CONSEQUENCES AFTER STORM AND TOTAL IMPACT ON TRANSMISSION NETWORK

	110 kV	220 kV	400 kV
Total No. of disconnected OHLs	51	7	1
Total No. of OHL outages due to lightning strikes	10	3	0
Total No. of damaged OHLs	14	5	3
Total No. of damaged OHL towers	40	5	6

Figures 4 shows one of damaged towers on area of TA ZG, while figure 5 one of damaged towers on TA OS.



Fig. 4 – Damages on OHL 110 kV Međurić – Daruvar located in TA ZG (tower no. 3) due to storm on July, 2023

Apart from the unavailability of a large number of OHLs, as a consequence of the two storms, there was no major loss of consumption (EnS<50 MWh), except during the moments of the storm in certain localities.



Fig. 5 – Damage on OHL 400 kV Ernestinovo – Sremska Mitrovica located in TA OS (tower no. 120) due to storm on July, 2023

In the event of a power line emergency, it is necessary to act immediately according to pre-arranged procedures, which requires organized work and clearly defined procedures with the aim of establishing the normal operating state of the power line as quickly as possible.

After the accident, the most important thing is to start looking for possible network power supplies, and then to perform urgent temporary or permanent rehabilitation of the damaged transmission line, based on the realistic possibilities in the network and considering the significance of the damaged power lines.

For that reason, modular towers are used. Emergency modular towers enable the establishment of temporary operation of a damaged overhead line in such a way that interruption of operation is minimal. The same is achieved in such a way that, after their installation, conductors and grounding wires are transferred to them. This enabled uninterrupted operation of the transmission line until the construction of new steel lattice tower, and the interruption of operation was limited only to the time needed to disconnect the conductors and grounding wires from the existing pole and hang them on temporary modular towers. Example of their implementation is shown on Figure 6, and previously used and elaborated in [7].



Fig. 6 – Example of modular tower installation

HOPS has modules and equipment for the formation of 110 kV, 220 kV and/or 400 kV towers [7], but it is important to mention that without support of neighbouring TSOs and loan of their modular towers, there wouldn't be enough of them and the normal operation of the transmission network would not be established.

III. Overview of Existing Overhead Lines Design Requirements

The existing transmission lines at HOPS are designed in accordance with the "Regulation on technical norms for the construction of overhead power lines with a nominal voltage of 1 kV to 400 kV" [9] (hereinafter - Regulation).

The wind load is equal to the product of the object's surface, wind pressure, the coefficient of wind action and the sine of the attack angle, where the horizontal wind direction is taken, and the wind load is perpendicular to the attack surface (Article 8, Regulation). When calculating the wind load, the surface of the object is considered as the actual surface, without additional load, attacked by the wind. For cylindrical columns and for grid columns, only the surfaces facing the wind are taken into account (Article 9, Regulation).

For almost all OHLs in the continental part of Croatia, under the jurisdiction of TAs ZG and OS, a basic wind pressure of 600 N/m^2 is assumed for dimensioning (in accordance with Article 10, Regulation). Also based on Art. 10 of the Regulation, the relationship between wind speed and pressure is defined:

$$P = \frac{v^2}{16}\ [daN/mm^2]$$

where v is the maximum wind speed [m/s] for the area which appears in the period of the last 5 years (or in the longer observed period for 400 kV lines).

The coefficients of wind action on individual parts of transmission lines are given in Art. 11 of the Regulations.

Therefore, the maximum wind speed for the calculation of wind pressure is determined on the basis of several years of measurements and the application of statistical processing of measurement data.

All new building structures should be dimensioned according to the Eurocodes, and for wind loads, the proposal known also as “NNA” (Croatian addition to the norm, that is still not approved) was created for HRN EN 1991-1-4:2012, Eurocode 1: Actions on structures - Part 1-4: General actions - Actions of the wind - National supplement, with a wind map for Croatia [10].

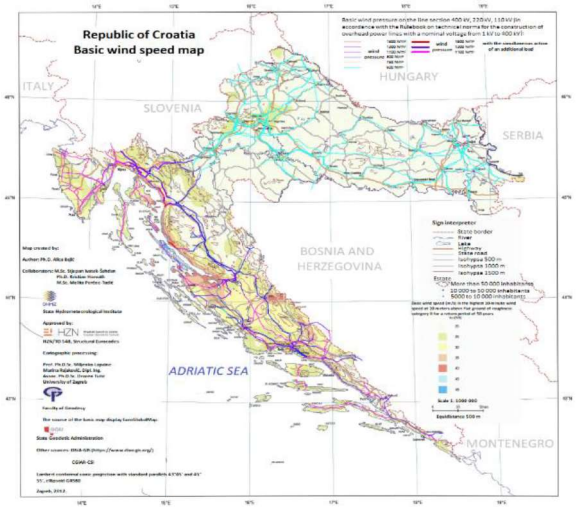


Fig. 9 – Map of the basic wind speed in the Republic of Croatia, with the routes of the existing OHLs and the wind pressure used during their design

As can be seen on the map, in the continental area the basic wind speed is 20-25 m/s, with maximal wind gust of 40 m/s and the probability that it will be exceeded in 50 years is 2%. On the coastal area, the basic wind speed exceeds 25 m/s, with maximal wind gust greater than 40 m/s. Even nowadays during design stage of OHLs (Eurocode), the loads on the structure are calculated differently than in the past, from previous chapter we saw that today is more and more possible to experience extreme weather conditions. It forces us to think weather the basic wind maps, which are an important input for the calculation at design stage, should be updated more often. Also, such maps should be correlated with most critical overhead lines from system operation state of view (according to N, N-1 and similar criteria), what could be trigger for TSO to require from OHL designers, consideration of stricter requirements.

IV. RECCOMENDATIONS AND LESSONS LEARNED

This paper shows example of direct impact of climate changes on essential infrastructure elements, such as power system overhead lines. It is shown that on areas with basic expected wind speed in range 20-25 m/s, a wind with strength above 50 m/s can occur and consequently, cause significant damage on wider area, and therefore endanger the safe operation of the entire power system. From this, we can conclude that TSOs should be prepared to act immediately, and therefore:

- ensure enough modular towers and related equipment,
- educate technical staff from their companies on how to design and mount such equipment,
- prepare the "action plan for unpredictable weather phenomena" with neighbouring TSOs/DSOs and other organisations in country,
- update weather data and prepare risk maps on yearly basis (i.e. wind, ice, snow, wildfire, floods, lightings, salt, dust and other possible extreme conditions),
- correlate such maps with most critical overhead lines from system operation state of view,
- establish working group that will recognize and recommend stricter requirements according to valid standards and communicate them to the transmission line designers (i.e. NNA in Croatia).

It is obvious that the causes of OHL outages can be different and that it is impossible to predict them, but the same principles apply for their repair.

V. CONCLUSION

Even if is known that more research is needed to statistically prove that climate is changing and directly affecting essential power systems, in this paper, it is shown that more and more frequent extreme weather conditions can occur in short time span. Therefore, various weather phenomena can affect power systems or concretely its essential parts such as overhead lines. Understanding these potential impacts is crucial for designing resilient and adaptive energy infrastructure.

Due to strong storms that occurred on July 19th and 21st, 2023 and direct impact of strong wind and lightings strikes, in total 59 OHLs were disconnected from the network. Almost half of these OHLs were significantly damaged. In total 51 transmission towers were damaged, what is equal to approximately 17 km of transmission route. In order to immediately restore the damaged part of transmission network, in total 22 modular emergency restoration towers are used. Other damages require more time for repair.

This fact led us to conclusion that detailed plan and preparation is necessary for such events, together with determination of few critical transmission lines that will likely remain undamaged, and as such help in restoring the network as soon as possible.

Also, TSO should have enough backup in terms of modular emergency restoration towers, based on unpredictable storm presented in this paper, to cover at least 20-25 km route of transmission lines.

Addressing the impacts of climate change on power systems requires a combination of strategies, including infrastructure improvements, the adoption of renewable energy sources, and the development of adaptive technologies. Planning for resilience and incorporating climate considerations into energy policies is essential to ensure the reliability and sustainability of power systems in the face of a changing climate.

Taking into account all the events of the last few years, it is evident that the impact of climate change is gaining momentum. Although the number of OHL outages and damages cannot be immediately visible in the statistics of operating events, it forces us to think, and at least to check the valid requirements, in order to adapt to the newly created circumstances. It can be beneficial for design of new OHLs, especially if they are located in areas sensitive on some of above mentioned climate factors.

Along with the current energy transition, a change in the entire paradigm of designing and building new energy facilities, especially when talking about new OHLs is necessary, already in the early stage of its development and consideration of new corridors.

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